HANQIN LYU

Cloud Security Engineer | Azure Security · SC-500 · SC-200 · SC-300 452 483 669 | Melbourne, VIC | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

Azure Cloud Security Engineer holding SC-500 (Microsoft Certified: Cloud and AI Security Engineer Associate), SC-200 (Microsoft Security Operations Analyst), SC-300, and CompTIA Security+, with hands-on experience in detection engineering, SIEM, cloud security, security automation, and infrastructure-as-code security. Experienced building detection content, automated response workflows, and security guardrails using Azure Sentinel, Defender for Cloud, Terraform, GitHub Actions, Azure Policy, and PowerShell/Python scripting. Proven track record translating security requirements and threat intelligence into practical, measurable controls across enterprise cloud environments. Clear written and verbal communication developed across APAC enterprise engagements. Bilingual (English/Mandarin).

Core Competencies: Detection Engineering (Sentinel/KQL) | SIEM & SOAR | Cloud Security (Azure/AWS) | DevSecOps & IaC Security (Terraform / Checkov / Azure Policy) | Security Automation (PowerShell/Python) | Threat Intelligence-Led Detection | Incident Response & Runbooks | Identity & Access Governance | Compliance (PCI DSS / NIST / ISO 27001) | Bilingual Communication (English/Mandarin)

WORK EXPERIENCE & PROJECT

QSA Associate (Internship) | Stratica

July 2025 – December 2025

Corporate advisory firm specialising in cloud risk, PCI DSS compliance, and strategic sourcing

- **IAM & Compliance Review**: Reviewed IAM policies, Entra ID conditional access configurations, and AWS access controls across cloud-hosted environments to verify PCI DSS compliance — identifying privilege misconfigurations and recommending least-privilege remediation steps
- **Client Technical Communication**: Delivered clear written and verbal updates to enterprise clients across APAC, translating complex technical findings into actionable remediation steps
- **Documentation & Access Reporting**: Prepared Attestations of Compliance (AOC), assessment reports, and bilingual (English/Mandarin) client materials — ensuring 100% accuracy across all deliverables
- **Stakeholder Collaboration**: Coordinated with senior consultants and client teams to track open issues, verify resolution, and maintain audit trails

Cloud Infrastructure Specialist (Project) | Melbourne Polytechnic

July 2024 – November 2024

AWS Network Environment Deployment — Academic Capstone

- **Infrastructure Deployment**: Architected and deployed a multi-tier AWS VPC with cross-AZ redundancy and auto-scaling, achieving 99.9% uptime and full CIS AWS Foundations Benchmark compliance
- **IAM & Access Control**: Designed and enforced IAM roles, SCPs, and least-privilege access policies across a multi-tier AWS environment; implemented RBAC and conditional access controls aligned with Zero Trust principles; configured VPN and Direct Connect for secure hybrid identity connectivity
- **Monitoring & Alerting**: Set up CloudWatch dashboards and alarms for real-time visibility into infrastructure health and performance
- **Incident Simulation & Remediation**: Identified and resolved 5 critical vulnerabilities through structured assessments; automated response workflows with Python/PowerShell, cutting response time by 35%

Network Technician (Project) | Melbourne Polytechnic

April 2024 – June 2024

Network Security Infrastructure Configuration

- **Network Design**: Deployed Cisco multilayer infrastructure with OSPF routing across 3 sites, maintaining 99.9% uptime
- **Security Hardening**: Configured zone-based firewalls, micro-segmentation, Cisco IPS/IDS (98% detection rate), and WPA3-Enterprise wireless with RADIUS authentication
- **Troubleshooting & Testing**: Conducted penetration testing and validated end-to-end connectivity and security controls post-deployment

Identity Governance Engineer (Personal Project) | Self-Directed

Jun 2026 – Jul 2026

Zero Trust identity governance architecture on Microsoft Entra ID, aligned to the SC-300 skills framework

- **Zero-Secret Identity Architecture**: Implemented Workload Identity Federation (OIDC) and Managed Identity for GitHub Actions and Azure Automation, eliminating long-lived client secrets across CI/CD and native Azure workloads
- **Privileged & Risk-Adaptive Access**: Converted high-privilege directory and Azure resource roles (Global Administrator, Key Vault, Automation Account) to time-bound PIM Eligible assignments; designed 6 layered Conditional Access policies enforcing phishing-resistant MFA for privileged roles and risk-based response for standard users
- **Identity Lifecycle Automation**: Built Entitlement Management access packages (time-bound and recurring-review models) and full Joiner-Mover-Leaver Lifecycle Workflows automating onboarding, department transfers, and offboarding with token revocation

Cloud Security Engineer (Personal Project) | Self-Directed *Sep 2026 – Nov 2026*

Shift-left DevSecOps pipeline and subscription-level Azure Policy guardrails for an Azure workload, aligned to the SC-500 skills framework

- **Infrastructure as Code & Pipeline**: Defined an Azure workload (SQL, Key Vault, immutable-retention storage, AI Services, VNet/NSG) from 6 Azure Verified Modules in Terraform (36 managed resources), deployed only through GitHub Actions with Entra-authenticated remote state (no shared keys)
- **Shift-Left Security Gates**: Built 3 pull-request gates (fmt/validate, Checkov, gitleaks) that must pass before plan; caught 2 of 2 deliberately seeded misconfigurations (SSH open to the internet, hardcoded SQL password) and fixed a scanner blind spot that had reported zero findings on AVM modules — 8 failed checks surfaced, triaged to 4 with documented reasons
- **Least-Privilege Delivery**: Split CI (Reader) and CD (Contributor) over GitHub OIDC with 0 stored client secrets; gated production apply behind a GitHub Environment with a required reviewer; CD identity cannot modify Azure Policy
- **Preventive Guardrails**: Authored a custom Azure Policy denying inbound SSH from any source, rolled out Audit-then-Deny at subscription scope; the same code deployed under Audit and was rejected with 403 RequestDisallowedByPolicy after CI gates were bypassed
- **AI-Assisted Engineering Governance**: Used Claude Code under a CLAUDE.md contract and 9 permission deny rules (no apply/destroy, no az login, no policy-module edits, no state reads), with no standing Azure credentials for the agent

EDUCATION & CERTIFICATIONS

Certificate IV in Cybersecurity | Melbourne Polytechnic
Master of Civil Engineering | Monash University

Professional Certifications:

- CompTIA Security+ — Network security, compliance, threat management, cryptography, risk management
- Microsoft SC-900 — Cloud security, Microsoft 365 Defender, Azure Sentinel, IAM, compliance frameworks
- SC-300: Microsoft Identity and Access Administrator — Identity governance, Entra ID, conditional access, privileged identity management
- SC-500: Microsoft Certified: Cloud and AI Security Engineer Associate — End-to-end Azure security, Defender for Cloud, Sentinel, AI workload security, network and storage security
- SC-200: Microsoft Security Operations Analyst Associate — Sentinel, Defender XDR, KQL, threat hunting

TECHNICAL SKILLS

Cloud Platforms & Services:

- AWS: VPC, EC2, IAM, Security Groups, NACLs, CloudWatch, Auto Scaling, Load Balancing, Direct Connect
- Azure: Entra ID (Azure AD), Conditional Access, Privileged Identity Management (PIM), Identity Governance & Lifecycle, App Registrations, SSPR, SAML / OAuth 2.0 / OIDC / SCIM, Azure Sentinel, Microsoft Defender for Cloud, Azure Policy, Azure Verified Modules

Monitoring, Networking & Security:

- SIEM/Monitoring: Splunk, ELK Stack, Azure Sentinel, CloudWatch, Wireshark, Sysmon
- Networking: Firewalls, IDS/IPS, VPN (IPsec/AES-256), OSPF, WPA3, RADIUS, DNS, DHCP
- Compliance & Frameworks: PCI DSS, ISO 27001, NIST 800-53, CIS Controls, SOC 2, OWASP Top 10

Scripting & Tools:

- Python, PowerShell, Bash — security automation, access provisioning scripts, incident response
- IaC & DevSecOps: Terraform, Bicep, GitHub Actions (OIDC), Checkov, gitleaks, Claude Code
- OS: Windows, Linux, MacOS | Version Control: Git/GitHub

Languages:

- English — Professional working proficiency
- Mandarin — Native proficiency

KEY ACHIEVEMENTS

- Designed and enforced least-privilege IAM policies and RBAC controls across AWS multi-tier VPC — achieving full CIS Benchmark compliance and 99.9% uptime
- 35–40% reduction in incident response time through Python/PowerShell automation
- 100% compliance accuracy across all PCI DSS deliverables at Stratica
- 98% attack detection rate in Cisco IPS/IDS configuration validated through penetration testing
- $100K+ cost savings identified via risk assessments during civil engineering role